

A Tool Bundle for AI Fairness in Practice

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FAIRNESS COMPASS

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FAIRNESS LIBRARY

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ABSTRACT

We developed two integrated tools to operationalise AI fairness in business practice: The *Fairness Compass* supports defining the fairness objective and thus specifies the metric. The *Fairness Library* helps selecting the best algorithm to achieve the adopted fairness metric. Together, both tools cover the entire implementation process of AI fairness.

CCS CONCEPTS

• **Computing methodologies** → **Philosophical/theoretical foundations of artificial intelligence.**

KEYWORDS

fairness and bias in artificial intelligence; ethics of AI

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1 INTRODUCTION

The last few years have seen a number of tremendous breakthroughs in the field of artificial intelligence (AI). A significant part of this success is due to major advances in machine learning (ML), the data analytics technique behind AI. Machine learning does recognise correlations in large data sets. Due to its ability to process loads of information in short time, it can uncover statistical patterns in data that humans cannot spot. This gives access to new kind of insights from the data which allow improved information analysis and model predictions. Although not absolutely error-free, the results clearly outperform conventional approaches, and often even

human experts. The areas of application are extensive and include medical diagnosis, university admission, loan allocation, recidivism prediction, recruitment, online advertisement, face recognition, language translation, recommendation engines, fraud detection, credit limits, pricing and false news detection.

The heavy dependence on data poses a new challenge though. The data used for training a ML algorithm are considered the *ground truth*. This means that during the learning phase, these data constitute the comprehensive representation of the real world which the algorithm seeks to approximate. If the training data includes any kind of unwanted bias, the resulting algorithm will incorporate and enforce it. Worse still, in the absence of robust explanations for the results, it is hardly possible for humans to recognise biased predictions of ML algorithms as such. Unwanted bias may happen to be directed against sensitive subgroups, defined for instance by gender, ethnicity or age. As a consequence, people from one such group would be generally disadvantaged by the system [2, 5, 7, 13].

2 PROBLEM

Fortunately, the research community has recognized the risk and plenty of mathematical definitions of fairness and many technical mitigation strategies have been proposed during the last years [9]. Nevertheless, the implementation of fair AI in production is still at an early stage.

One reason is that the importance of the careful choice of fairness objective is often overlooked. While the available definitions reflect existing, widespread notions of fairness in society, they also do encode quite different objectives. Satisfying them all at the same time was shown to be impossible [3, 6, 8]. The best option therefore depends on the use case and its context, and it is a matter of ethical criteria and legal requirements. Many research proposals, however, focus on the technical aspect of the mitigation approach, and the kind of fairness they support plays a minor role. Technical frameworks for AI fairness assess biases with respect to all sorts of fairness metrics, but they provide little orientation on how to make the right choice [1, 4]. In fact, some mainly highlight the trade-offs between fairness and accuracy for each metric, which is not a sound basis for a sustainable decision-making in this matter.

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Implementing fairness in production is a multi-disciplinary effort where certain decisions have to be made by professionals with different skills and expertise. This includes business owners, lawyers, ethics advisors, data scientists, data engineers, and others. While bringing together all those actors often is a complex process, there is no way around it: If the choice of the fairness objective is not well-considered, or its enforcement is not implemented with utmost care, the fairness-enhancing intervention may have unintended effects. If the intention of the implemented fairness definition is not clearly communicated, disappointment is inevitable when the stakeholders' inner expectations differ.

3 PROPOSAL

We propose a workbook for AI practitioners which covers the full process of implementing fair AI in practice. It consists of two integrated components: First, the *Fairness Compass* to specify the fairness objective for the given application context. Second, the *Fairness Library* to choose the technical approach which supports the smoothest integration into the existing AI implementation.

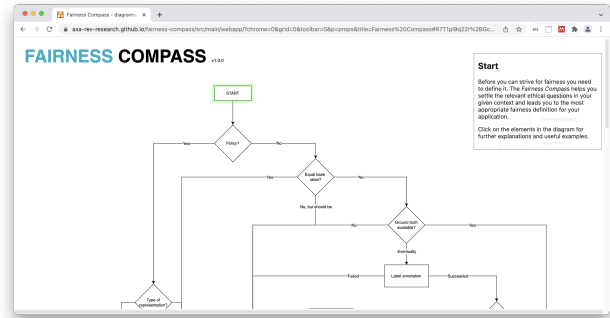
3.1 Fairness Compass

The *Fairness Compass* is a tool to identify the most appropriate fairness definition for a given AI system [10–12]. At its core, it structures the complex landscape of fairness metrics by their different properties. The resulting schema is a decision tree which formalises the decision process. There are three different types of nodes in the diagram: The diamonds symbolise decision points, the white boxes stand for actions and the grey boxes with round corners are the fairness definitions. The arrows which connect the nodes represent the possible choices. Based on the answers to a few questions around existing policies, the quality of the data, the potential impact of the application and others, the *Fairness Compass* will provide the most suitable fairness metric for the described scenario. For increased usability, the schema was implemented as interactive online tool. In this version, tooltips with extended information, examples and references facilitate navigating the tree (see Figure 1a).

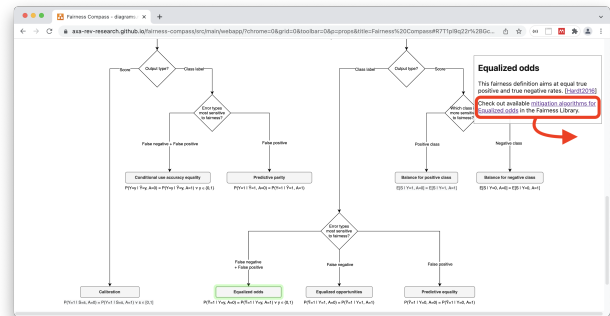
The user continues the process until the leaf node of the decision tree is reached. This end node shows the final outcome of the decision path. It represents the most appropriate fairness definition for the described application scenario. The node provides a deep link to the *Fairness Library* (see Figure 1b).

3.2 Fairness Library

The *Fairness Library* is a collection of algorithms for bias mitigation. It provides the short descriptions of available research papers in this field, including links to the full article and their source code. Most importantly, it arranges the different proposals by supported fairness metric. This way technical stakeholders can filter all algorithms for the previously decided fairness definition and get a quick overview of the available options. Choosing the best solution for their existing system environment is now a straightforward process.



(a) Questions about the application scenario guide the user to the most appropriate fairness definition.



(b) The result links to available implementation approaches for the identified fairness definition.

Figure 1: The *Fairness Compass* as interactive tool (online).

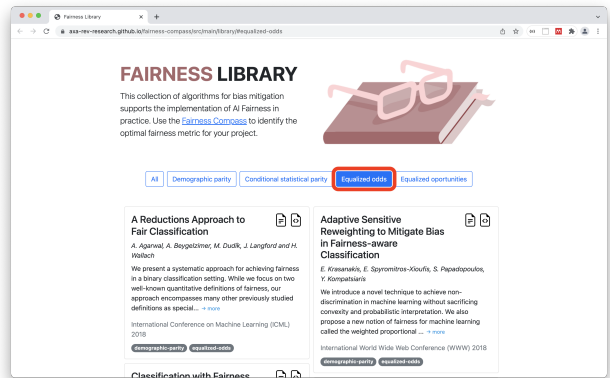


Figure 2: The *Fairness Library* aggregates the available fairness mitigation approaches per supported fairness metric (online).

4 CONCLUSION

We present a one-stop shop for operationalizing responsible AI in practice. Different from other proposals in this area, we put the choice of the fairness objective first. The *Fairness Compass* provides business owners a straightforward, transparent procedure to settle for the most appropriate fairness definition for a given application scenario. The *Fairness Library* helps technical staff to choose from the available implementation options.

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